

PROBLEMS CLASS 1: SOLUTIONS

1. A simple epidemic model

Consider a population which is either susceptible (S) or infected (I) with a disease.

- (a) We assume that the total population, N , is constant. Given the population is either susceptible or infected, write down a simple equation for N .
- (b) Infected people infect susceptible people by interacting with them. By considering how we represented interactions between greenflies and ladybirds, write down an equation for dI/dt in terms of I and S .
- (c) By substituting in N , eliminate S from this equation and show that

$$\frac{dI}{dt} = \beta I(N - I). \quad (1)$$

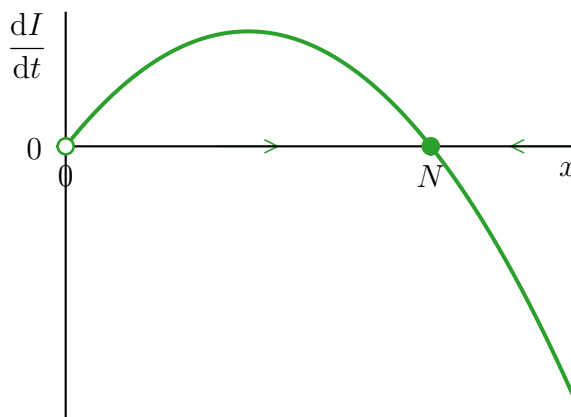
Do you recognise this type of model from chapter 1?

- (d) Find the equilibria of this model for I , establish when they are permissible, and assess their stability. You can assess the stability either graphically, or by doing an algebraic linear stability analysis (with ε etc.)
- (e) What does this mean for the spread of the disease?

Solution:

- (a) $N = S + I$
- (b) $dI/dt = \beta IS$, where $\beta > 0$.
- (c) $dI/dt = \beta I(N - I)$. This is a logistic model.
- (d) The equilibria are where $dI/dt = 0$, i.e. $I = 0, N$ (no-one or everyone).
 To assess their stability, we can use two methods.

Method I: Graphical analysis Graphical analysis of the phase plane is really convenient, but often it requires us to make assumptions about the signs of our constants which we have to be careful to be explicit about. This isn't something to catch you out! Look here, where we've plotted the phase plane (dI/dt against I):



This is the shape of the graph assuming $\beta > 0$ and $N > 0$. These are part of our model choices, so that's OK. The graph allows us to see instantly that 0 is unstable, and N is stable.

Method II: Formal linear stability analysis Let $I = I_0 + \varepsilon I_1$. Then

$$\varepsilon \frac{dI_1}{dt} = \beta(I_0 + \varepsilon I_1)(N - I_0 - \varepsilon I_1). \quad (2)$$

For $I_0 = 0$:

$$\varepsilon \frac{dI_1}{dt} = \beta \varepsilon I_1 N \quad (3)$$

to order ε , giving us

$$I_1 = A e^{\beta N t}. \quad (4)$$

Since $\beta N > 0$, this is unstable.

For $I_0 = N$:

$$\varepsilon \frac{dI_1}{dt} = -\beta N \varepsilon I_1 \quad (5)$$

to order ε , which is asymptotically stable. Given that this is a logistic map, we sort of knew this anyway.

- (e) What this means for the disease is that it spreads without bound so long as $\beta > 0$.

2. A slightly less simple epidemic model

More realistic is to keep what we have so far, but also to allow a proportion of the infected population to recover, after which they become susceptible again. We continue to assume that N is constant, so the same equation for N from question 1 applies.

- (a) Write down a new equation for dI/dt in terms of I and S . Once again substitute in N to remove S from this equation, and show that

$$\frac{dI}{dt} = I[\beta(N - I) - \gamma]. \quad (6)$$

- (b) Find the equilibria of this model for I , establish when they are permissible, and assess their stability. Show that the disease spreads if $R_0 > 1$, where R_0 is a ratio of terms you will find.
- (c) By considering the units of the terms in the definition of R_0 , convince yourself that it is dimensionless.

Congratulations! You have found our first definition of the famous R_0 . This term is known as the basic reproductive ratio, and as we heard plenty of in the news during the peak of the pandemic crisis, can be interpreted as the average number of susceptible people that an infected person infects in a disease-free population of size N .

Solution:

(a) This time,

$$\frac{dI}{dt} = \beta IS - \gamma I \quad (7)$$

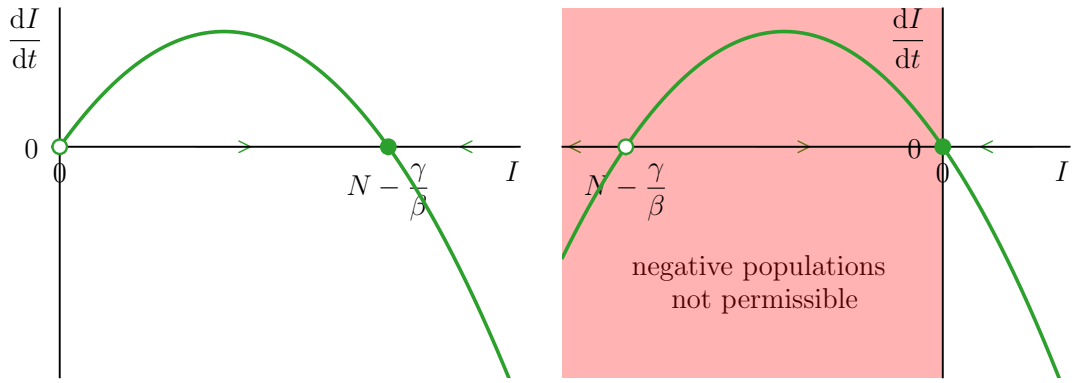
$$= I(\beta(N - I) - \gamma), \quad (8)$$

where $\beta, \gamma > 0$.

(b) The equilibria are where $dI/dt = 0$, i.e. $I = 0, N - \gamma/\beta$. The second equilibrium is permissible if $N - \gamma/\beta > 0$, i.e. $\beta N/\gamma > 1$.

To assess their stability, we once again have two methods.

Method I: Graphical analysis Once again we can plot the phase plane: dI/dt v I . Let's be careful, because a priori either of these phase planes are possible, depending on whether $N - \gamma/\beta \gtrless 0$:



We can see:

- If $N - \gamma/\beta > 0$, then the graphical analysis says both equilibria are permissible; $I = 0$ is unstable; and $I = N - \gamma/\beta$ is stable.
- If $N - \gamma/\beta < 0$, then only the zero equilibrium is permissible, and the graphical analysis (which still works for negative populations, even if it is meaningless in the real-world context of the model) tells us $I = 0$ is stable.

Introducing $R_0 = \beta N/\gamma$, this gives us:

- If $R_0 > 1$, there are two equilibria and the disease spreads (away from $I = 0$).
- If $R_0 < 1$, only the zero-infection equilibrium exists and the disease dies out.

Method II: Formal linear stability analysis

$$\varepsilon \frac{dI_1}{dt} = (I_0 + \varepsilon I_1) [\beta(N - I_0 - \varepsilon I_1) - \gamma]. \quad (9)$$

For $I_0 = 0$:

$$\varepsilon \frac{dI_1}{dt} = \varepsilon I_1 [\beta N - \gamma] \quad (10)$$

to order ε , giving us

$$I_1 = A e^{(\beta N - \gamma)t}. \quad (11)$$

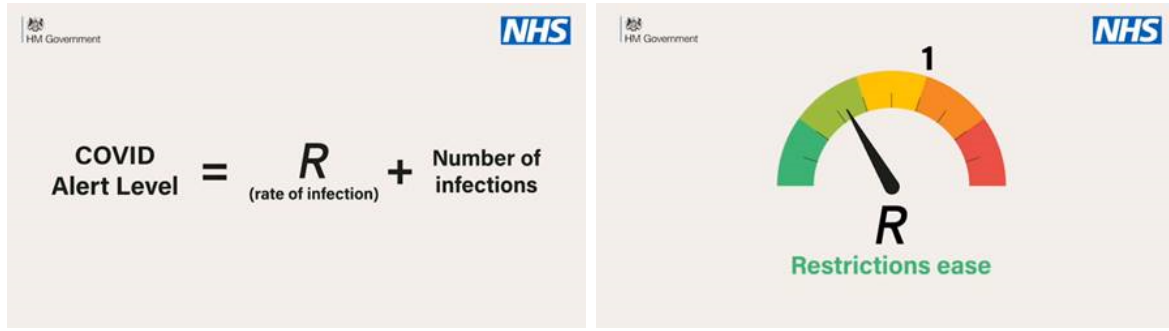


Figure 1: Government slides from coronavirus briefing, 10 May 2020

This is unstable if $\beta N - \gamma > 0$, i.e. $R_0 = \beta N / \gamma > 1$.

For $I_0 = N - \gamma / \beta$,

$$\varepsilon \frac{dI_1}{dt} = -\varepsilon I_1 [\beta N - \gamma] \quad (12)$$

to order ε , giving us

$$I_1 = A e^{-(\beta N - \gamma)t}. \quad (13)$$

If $R_0 > 1$ (also the condition for this equilibrium to be permissible), this is stable.

We can conclude that:

- If $R_0 > 1$, there are two equilibria and the disease spreads (away from $I = 0$).
 - If $R_0 < 1$, only the zero-infection equilibrium exists and the disease dies out.
- (c) $R_0 = \beta N / \gamma$. Looking at our equation for dI/dt , which has units $1/T$ (for T time), γ must have units $1/T$ and β units of inverse time and density (or size). Finally since N is the total population size or density we see we have no units for R_0 .

3. Quick sanity check

Take a look at the government slides from May 2020 in fig. 1, where R_0 is represented by R . Given what we have just learnt about R_0 , make a quick critique of these slides.

Solution: Some obvious things here: it makes no sense to add R_0 to the number of infections, and the dial on the right suggests that there is a maximum and minimum R_0 .

4. The SIR model

We now move to the popular SIR disease model. This model separates the recovered population into a new group, R , assuming that they have become immune and therefore are no longer susceptible. The total population remains constant.

- (a) Write down equations for dS/dt , dI/dt , dR/dt and N .
- (b) So far, we haven't added birth and natural (non-epidemic) death to our model (N has been constant). In what sort of situation might this be a reasonable choice?

- (c) Given your equation for N , confirm that in your model, the total population is constant (i.e. that $dN/dt = 0$).
- (d) Find all equilibria of the system. Is it/are they asymptotically stable? (Hint: pay close attention to our three possible cases at the end of section 2.2.)
- (e) By combining two of your equations, find S as a function of R and hence demonstrate that this model predicts some individuals will always avoid infection if $S(0) > 0$.
- (f) Solutions for the infected population, I , either show monotonic decay or rise and have a peak infection value before decaying to the equilibrium $I = 0$. Demonstrate that the criteria for the disease reaching a peak, i.e. that I ever rises, is that

$$R_0 > \frac{N}{S(0)}. \quad (14)$$

Solution:

- (a) We have

$$\frac{dS}{dt} = -\beta IS, \quad (15)$$

$$\frac{dI}{dt} = \beta IS - \gamma I, \quad (16)$$

$$\frac{dR}{dt} = \gamma I, \quad (17)$$

$$N = S + I + R. \quad (18)$$

- (b) This choice is reasonable if the epidemic moves quite quickly.
- (c) Adding the derivatives,

$$\frac{dN}{dt} = \frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = -\beta IS + \beta IS - \gamma I + \gamma I = 0. \quad (19)$$

- (d) The equilibria ($dS/dt = dI/dt = dR/dt = 0$) only happen when $I = 0$. In which case, $S = S_0$, frankly anything, and $R = N - S_0$. So we see there is a one-parameter family of equilibria, $(S_0, 0, N - S_0)$, where $S_0 \in [0, N]$. It cannot be asymptotically stable as there is a nearby equilibrium always (it is a one-parameter family of equilibria). This is listed under the degenerate case in the notes.
- (e) We divide the rate equations for S and R to obtain

$$\frac{dS}{dR} = \frac{-\beta S}{\gamma}. \quad (20)$$

The solution is clearly an exponential,

$$S(t) = S(0)e^{-(\beta/\gamma)R} = S(0)e^{-R_0 R/N} \geq S(0)e^{-R_0}. \quad (21)$$

Now, since R_0 is a positive constant, $S(t)$ is bounded from below. The smaller R_0 , the larger this bound is as a fraction of the initial population (which makes sense since smaller R_0 means people are recovering relatively quickly compared to the rate of infection).

(f) Take the equation for I , which can be written as

$$\frac{dI}{dt} = I[\beta S - \gamma]. \quad (22)$$

Note that, since S is monotonically decreasing, the maximum value this rate can take is at $t = 0$. Thus for the solution to rise we would need

$$\beta S(0) - \gamma > 0, \quad (23)$$

giving the required condition.